NM2 - cv4

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1 Lorenzovy rovnice

$$\dot{u}_1 = \sigma(u_2 - u_1)$$
$$\dot{u}_2 = \rho u_1 - u_2 + u_1 u_3$$
$$\dot{u}_3 = -\beta u_3 + u_1 u_2$$

- Interval:$[0, 100]$
- Integrační krok: 10^{-2}
- Eulerova metoda

1.1 Změna v parametru β

Table 1: Analýza v parametru β

σ	ρ	β	$u_1(0)$	$u_2(0)$	$u_3(0)$
1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	6.0	1.0	1.0	1.0
1.0	1.0	11.0	1.0	1.0	1.0
1.0	1.0	16.0	1.0	1.0	1.0
1.0	1.0	21.0	1.0	1.0	1.0

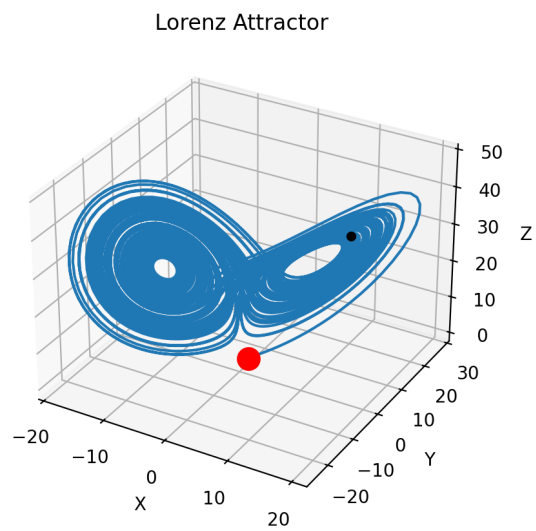

Figure 1: $\beta = 1$

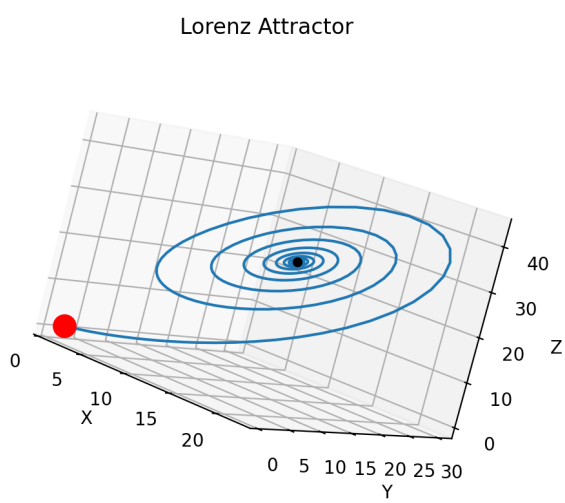

Figure 2: $\beta = 6$

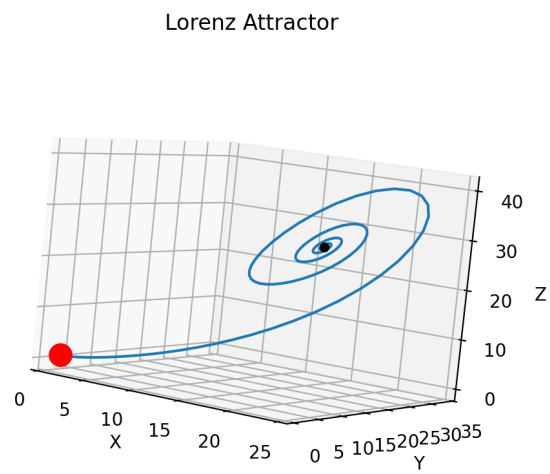

Figure 3: $\beta = 11$

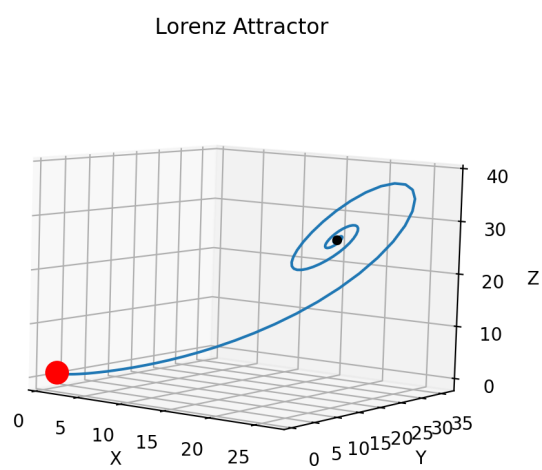

Figure 4: $\beta = 16$

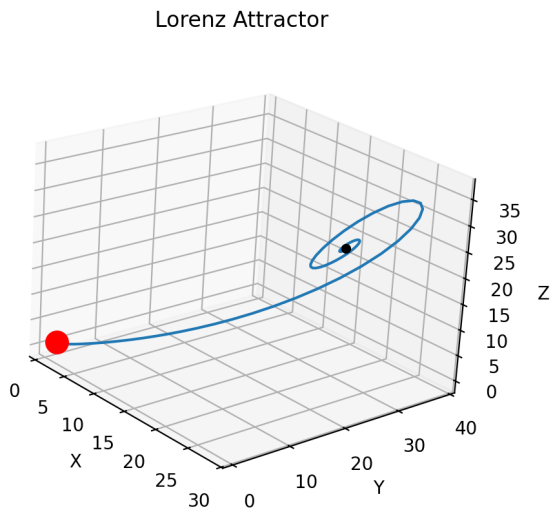

Figure 5: $\beta = 21$

1.2 Změna v počátečních podmínkách

Table 2: Analýza v počátečních podmínkách

σ	ρ	β	$u_1(0)$	$u_2(0)$	$u_3(0)$
1.0	1.0	1.0	1.0	1.0	1.0
1.0	1.0	1.0	31.0	31.0	31.0
1.0	1.0	1.0	61.0	61.0	61.0
1.0	1.0	1.0	91.0	91.0	91.0
1.0	1.0	1.0	121.0	121.0	121.0

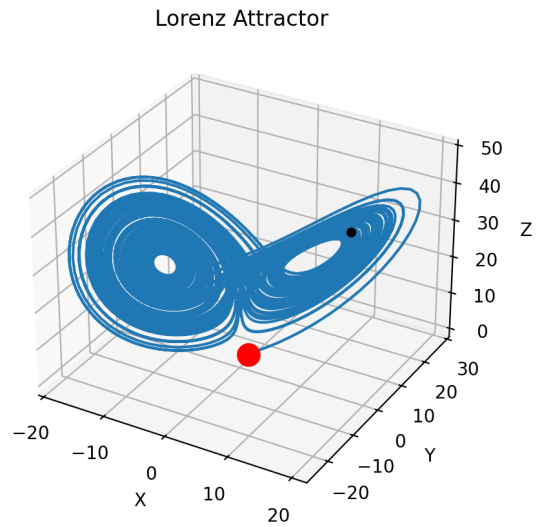

Figure 6: $u_1(0) = 1, u_2(0) = 1, u_3(0) = 1$

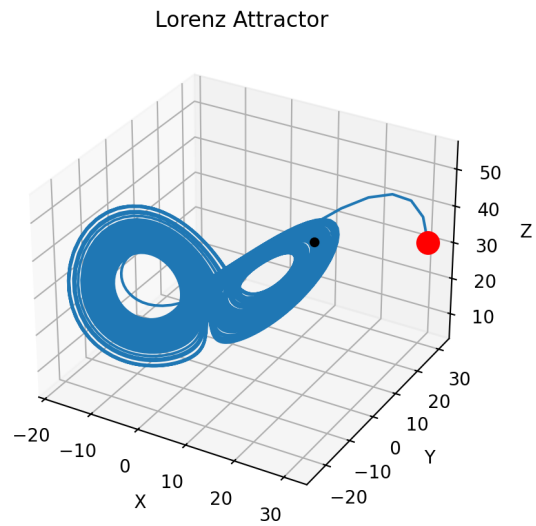

Figure 7: $u_1(0) = 31, u_2(0) = 31, u_3(0) = 31$

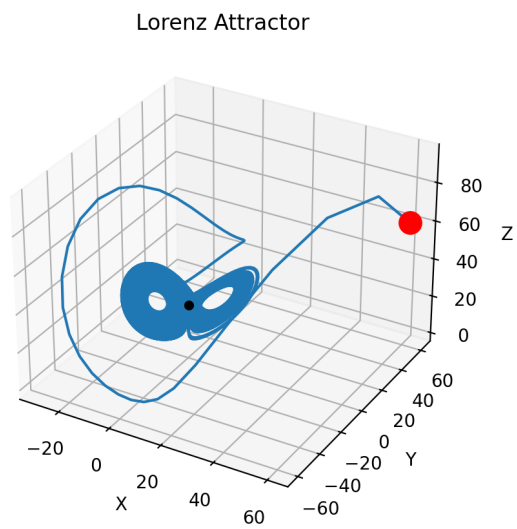

Figure 8: $u_1(0) = 61, u_2(0) = 61, u_3(0) = 61$

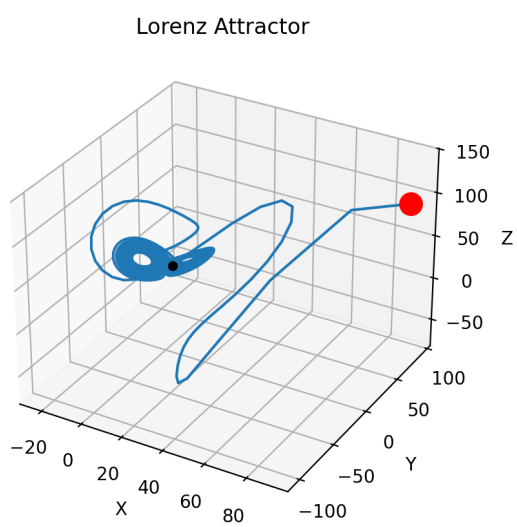

Figure 9: $u_1(0) = 91, u_2(0) = 91, u_3(0) = 91$

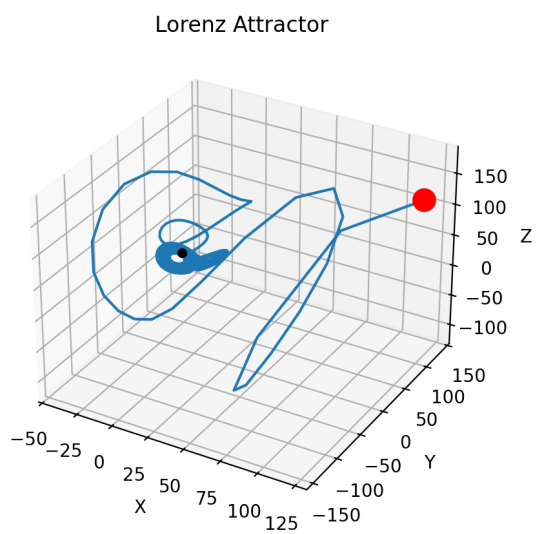

Figure 10: $u_1(0) = 121, u_2(0) = 121, u_3(0) = 121$